Simply Supported Beam Equilibrium and Response

CIVIL-ENGINEERING · civil-engineering-structural-engineering · BoardPrep lesson v1

Official examination scope

PRC's Civil Engineering board page provides official syllabus and examination references. This lesson practices basic equilibrium and elastic-beam relationships within that scope; confirm the current PRC notice and TOS. Formulas assume the stated idealized supports and model. No frequency claim is made.

Learning objectives

- Solve support reactions using a free-body diagram and force and moment equilibrium.
- Relate shear and bending moment to locate a candidate maximum moment.
- Compute midspan deflection for a stated simply supported beam model with consistent units.

1. Draw the free-body diagram

Replace supports with reaction components, show each load and its location, and choose positive directions. For planar statics, sum horizontal forces, vertical forces, and moments to zero. A centered load on a symmetric span gives equal reactions; a different geometry may not.

2. Use shear and moment together

Under the usual beam sign convention, dM/dx = V. A stationary moment occurs where shear is zero or changes sign, subject to jumps and boundary conditions. Check moment values at supports and other candidates before naming the maximum.

3. Apply the centered point-load model

For a simply supported prismatic beam, linear elasticity, small deflection, and a centered load P, delta_max = P L^3/(48 E I). Use P in newtons, L in metres, E in pascals, and I in m^4; the answer is metres.

Find support reactions

Question: A simply supported beam carries a centered 16 kN point load.
1. Vertical equilibrium: R_A + R_B = 16 kN.
2. Moments about A: R_B L = 16(L/2), so R_B = 8 kN.
3. Then R_A = 16 - 8 = 8 kN.
Answer: Each support reaction is 8 kN upward.
Check: The reactions sum to the load and their moments balance it.

Check midspan deflection

Question: P = 8 kN, L = 4 m, E = 200 GPa, I = 3.0 x 10^-5 m^4.
1. Convert P = 8000 N and E = 200 x 10^9 Pa.
2. Substitute: 8000(4^3)/(48(200 x 10^9)(3.0 x 10^-5)) = 0.00178 m.
3. Multiply by 1000 to convert to millimetres.
Answer: The deflection magnitude is approximately 1.78 mm.
Check: All inputs use consistent SI units.

Common misconceptions

- Assuming equal reactions when geometry or loading is asymmetric.
- Locating maximum moment without checking shear or the moment function.
- Mixing millimetres, pascals, metres, and m^4 without conversion.
- Applying a small-deflection formula outside its assumptions.

Practice patterns

- Find reactions from force and moment equilibrium.
- Locate the maximum moment from a shear or moment relationship.
- Apply a deflection formula and convert the result to the requested unit.

Official scope references

Official scope: PRC Civil Engineering Board and Syllabus Resources — Professional Regulation Commission; Professional Regulatory Board of Civil Engineering — https://prc.gov.ph/Pages/PRBv4/CivilEngineering.htm

Instructional references

Instructional reference: University Physics Volume 1, Section 12.1: Conditions for Static Equilibrium — OpenStax, Rice University — https://openstax.org/books/university-physics-volume-1/pages/12-1-conditions-for-static-equilibrium

Instructional reference: Analysis of Structural Member Systems — Jerome J. Connor; MIT OpenCourseWare — https://ocw.mit.edu/courses/1-571-structural-analysis-and-control-spring-2004/resources/open-textbooks/

Instructional reference: MIT OpenCourseWare 1.050 Solid Mechanics: Internal Forces and Moments, Exercise 3.16 — MIT OpenCourseWare, Department of Civil and Environmental Engineering — https://live.ocw.mit.edu/courses/1-050-solid-mechanics-fall-2004/5070a932644d2cb44cc965ac56f79c66_emech3_04.pdf

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